S4 Supplementary: Related Work

NDT in relation to dynamical-systems/landscape accounts, predictive-processing/active-inference frameworks, and integrated-information/network-control approaches

Article type: Supplement
Short title: S4 Related Work - NDT

Robin Langell[1]
[1] Langell Konsult AB, Vallentuna, Sweden
For correspondence: Robin Langell, hello@noeticdiffusion.com

Relation to the main article

This supplement expands the **Related work** appendix pointer from the main article (`NoeticDiffusionTheory.typ`, Appendix).

S4.0 Related work

NDT is closest to dynamical-systems and landscape accounts of brain activity, where cognition is modeled as movement through metastable state spaces (Deco et al., 2011; Kelso, 2012). Its distinctive addition is a rhythmic variance schedule: instead of treating noise as constant or exogenous, NDT asks whether neural rhythms provide an observable control signal that gates exploration and denoising over short time windows.

Predictive-processing and active-inference accounts provide an important neighboring framework (Clark, 2013; Friston, 2010; Seth, 2013). NDT is compatible with these accounts but shifts the emphasis from content-level prediction to trajectory geometry, variance timing, and empirical state-space readouts. Integrated-information and network-control approaches are also complementary: they foreground information structure or transition energy, whereas NDT focuses on event-locked flow, topology, recurrence, and reachability in measured neural trajectories (Gu et al., 2015; Tononi et al., 2016).

The comparison should be read conservatively. NDT does not claim to replace these frameworks, nor to solve the hard problem of consciousness. Its proposed contribution is a narrower measurement hypothesis: if rhythmic variance control is a real organizing principle, then event-locked entropy signatures and MNPS geometry should survive preregistered null models better than smooth-precision or rhythm-free alternatives.

S4.1 Companion-evidence ledgers (clinical and rhythm-related)

Alongside the theoretical comparison above, four internal evidence-survey ledgers catalog which companion manuscripts under `articles/CLINICAL/`, `articles/RVC_SLEEP*`, `articles/RichSLeep_EAP_v2/`, and related folders can serve as **bounded, dataset-level** empirical material for Signature Prediction 4 (clinical geometry contrasts) and the RVC/EAP rhythm claims. Each ledger follows the project's five-way evidence discipline (established / internal validated / plausible / speculative / rejected) and explicitly separates positive findings from negative and null results.

- **Coma/cardiac-arrest geometry** — `articles/NDT/parts/007_20260809_clinical_coma_evidence_survey.md`. Pooled reachability/MNJ contrasts between good and poor neurological outcome are internal validated, dataset-bounded associations; they are **not** a test of the main article's Grand Mal boundary note and do not validate MNDM, RVC, or TRN mechanics.
- **Non-coma clinical disorders** (psychosis, depression, Parkinson, Alzheimer/FTD) — `articles/NDT/parts/008_20260809_clinical_disorder_evidence_survey.md`. Medication-stratified psychosis geometry and BDI-linked depression reachability support Signature Prediction 4′s **existence** claim, but do not confirm the illustrative Mapper/DET or $d > 0.6$ volume-collapse operationalizations as literally written; several companions report informative null results.
- **Sleep rhythms and autonomic phase-locking** — `articles/NDT/parts/009_20260809_sleep_rhythm_evidence_survey.md`. Spindle-locked MNPS reconfiguration and carrier-specific autonomic phase-locking (SO trough $>$ spindle peak $>$ spindle onset $\approx$ REM-theta null) are internal validated; none of these companions validate the explicit $\hat{r}(t)$ construction from the main article's RVC section or a TRN causal mechanism.
- **EAP, development, and task geometry** — `articles/NDT/parts/010_20260809_eap_developmental_task_evidence_survey.md`. Wake EAP co-modulation, encoding-pace task geometry, and developmental MNPS trajectories are internal validated at the condition level, with explicit nulls for direct within-subject anchor-to-performance coupling and for age-related MNJ/reachability change.

These ledgers are maintained as living documents; editors updating Signature Prediction 4 or the RVC/EAP sections should consult them before adding or strengthening any clinical or rhythm-linked citation.

Bibliography

Clark, A. (2013). Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, *36*(3), 181–204.

Deco, G., Jirsa, V. K., & McIntosh, A. R. (2011). The dynamic brain: from spiking neurons to neural masses and cortical fields. *Plos Computational Biology*, *7*(1), e1002052.

Friston, K. (2010). The free-energy principle: a unified brain theory?. *Nature Reviews Neuroscience*, *11*(2), 127–138.

Gu, S., Pasqualetti, F., Cieslak, M., Telesford, Q. K., Yu, Q., Kahn, A. E., Medaglia, J. D., Vettel, J. M., Miller, M. B., Grafton, S. T., & Bassett, D. S. (2015). Controllability of structural brain networks. *Nature Communications*, *6*, 8414.

Kelso, J. A. S. (2012). Multistability and metastability: understanding dynamic coordination in the brain. *Philosophical Transactions of the Royal Society B*, *367*(1591), 906–918.

Seth, A. K. (2013). Interoceptive inference, emotion, and the embodied self. *Trends in Cognitive Sciences*, *17*(11), 565–573.

Tononi, G., Boly, M., Massimini, M., & Koch, C. (2016). Integrated information theory: from consciousness to its physical substrate. *Nature Reviews Neuroscience*, *17*(7), 450–461.